

Matrices and Arrays

Array creation, combining, reshaping, rearranging, and indexing

Matrices and arrays are the fundamental representation of information and data in MATLAB®. You can create common arrays and grids, combine existing arrays, manipulate an array's shape and content, and use indexing to access array elements. For an overview of matrix and array manipulation, watch [Working with Arrays](#).

Functions

[collapse all](#)

Create and Combine Arrays

zeros	Create array of all zeros
ones	Create array of all ones
rand	Uniformly distributed random numbers
true	Logical 1 (true)
false	Logical 0 (false)
eye	Identity matrix
diag	Create diagonal matrix or get diagonal elements of matrix
blkdiag	Block diagonal matrix
cat	Concatenate arrays
horzcat	Concatenate arrays horizontally
vertcat	Concatenate arrays vertically
repelem	Repeat copies of array elements
repmat	Repeat copies of array
combinations	Generate all element combinations of arrays (<i>Since R2023a</i>)

Create Grids

linspace	Generate linearly spaced vector
logspace	Generate logarithmically spaced vector
freqspace	Frequency spacing for frequency response

<code>meshgrid</code>	2-D and 3-D grids
<code>ndgrid</code>	Rectangular grid in N-D space

Determine Size, Shape, and Order

<code>length</code>	Length of largest array dimension
<code>size</code>	Array size
<code>ndims</code>	Number of array dimensions
<code>numel</code>	Number of array elements
<code>isscalar</code>	Determine whether input is scalar
<code>isvector</code>	Determine whether input is vector
<code>ismatrix</code>	Determine whether input is matrix
<code>isrow</code>	Determine if input is row vector
<code>iscolumn</code>	Determine if input is column vector
<code>isempty</code>	Determine whether array is empty
<code>issorted</code>	Determine if array is sorted
<code>issortedrows</code>	Determine if matrix or table rows are sorted
<code>isuniform</code>	Determine if vector is uniformly spaced (<i>Since R2022b</i>)

Resize, Reshape, and Rearrange

Resize

<code>head</code>	Get top rows of array or table
<code>tail</code>	Get bottom rows of array or table
<code>resize</code>	Resize data by adding or removing elements (<i>Since R2023b</i>)
<code>paddata</code>	Pad data by adding elements (<i>Since R2023b</i>)
<code>trimdata</code>	Trim data by removing elements (<i>Since R2023b</i>)

Reshape

<code>permute</code>	Permute array dimensions
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<code>ipermute</code>	Inverse permute array dimensions
<code>shiftdim</code>	Shift array dimensions
<code>reshape</code>	Reshape array by rearranging existing elements
<code>squeeze</code>	Remove dimensions of length 1

Rearrange

<code>sort</code>	Sort array elements
<code>sortrows</code>	Sort rows of matrix or table
<code>flip</code>	Flip order of elements
<code>fliplr</code>	Flip array left to right
<code>flipud</code>	Flip array up to down
<code>rot90</code>	Rotate array 90 degrees
<code>transpose</code>	Transpose vector or matrix
<code>ctranspose</code>	Complex conjugate transpose
<code>circshift</code>	Shift array circularly

Indexing

<code>colon</code>	Vector creation, array subscripting, and for-loop iteration
<code>end</code>	Terminate block of code or indicate last array index
<code>ind2sub</code>	Convert linear indices to subscripts
<code>sub2ind</code>	Convert subscripts to linear indices

Topics

Creating, Concatenating, and Expanding Matrices

Create a matrix or construct one from other matrices.

Array Indexing

Access elements of an array by specifying their indices or by checking whether elements meet a condition.

- [Find Array Elements That Meet Conditions](#)
- [Access Data in Cell Array](#)
- [Access Data in Tables](#)
- [Structure Arrays](#)

- [Comma-Separated Lists](#)
- [Indexing into Function Call Results](#)
- [Detailed Rules for Array Indexing](#)

Indexed Assignment

Use indexed assignment to modify array elements and enable array growth.

- [Arrays with Dimensions of Size 1](#)
- [Detailed Rules for Indexed Assignment](#)

Removing Rows or Columns from a Matrix

Remove matrix rows or columns.

Reshaping and Rearranging Arrays

Change the shape or arrangement of elements in an existing array.

Multidimensional Arrays

Create and manipulate arrays with three or more dimensions.

Empty Arrays

Create and use empty arrays, which have no elements but have specific dimensions, at least one of which is 0.

Teaching Resources

Fundamentals of Programming

Learn the underlying concepts of code, data, arrays, branching, loops, and algorithms.

YOU MIGHT ALSO BE INTERESTED IN

